



The Reward Line

A Matched-Flyer Field Trial of Tip-Off Call Volume and an Audit of the Neighbourhood Noticeboard Reward-Tier Schedule

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Abstract. Neighbourhood noticeboards are governed, at this University, by a Reward-Tier Schedule: four reward amounts (\$20, \$50, \$100, \$200) mapped to the geographic reach a lost-pet flyer is expected to need, on the assumption that a larger stated reward reliably purchases proportionally more tip-off calls. We report a six-week matched-flyer field trial across 35 residential streets, each carrying a paired posting of an identical lost-cat flyer differing only in whether a reward was stated and, where one was, at which of the Schedule's four tiers. A shared intercept line, keyed to a per-flyer reference code, logged every incoming call. Flyers stating any reward received at least one call in 39.0% of street-weeks against 23.8% for otherwise identical no-reward flyers, a 15.2 percentage-point gap (95% CI 6.4-24.0). Among reward flyers, mean call volume rose from the \$20 to the \$50 tier and then held flat through \$100 and \$200; the Schedule's own predicted curve, which continues climbing steeply across the same range, diverges sharply from the observed data above \$50. An ablation comparing the Schedule's two highest tiers directly found no reliable difference in call volume between them. We discuss what a schedule calibrated to a curve that does not hold past its midpoint is entitled to keep recommending.

Introduction

A lost-pet flyer is among the oldest pieces of informal neighbourhood signage: a photograph, a description, a phone number, stapled to a power pole or pinned to a community noticeboard by someone who wants their animal back. Whether and how much reward to offer is usually left to the person doing the stapling. At this University, it is not. The Adaptive Metrics Lab's Neighbourhood Noticeboard Reward-Tier Schedule assigns a recommended reward amount to each of four geographic-reach tiers — \$20 for a single-street posting run, \$50 for a multi-street run, \$100 for a suburb-wide run, and \$200 for a run crossing into an adjacent suburb or an online cross-post — on the working assumption that a larger stated reward reliably buys a proportionally larger volume of tip-off calls across the whole range. The Schedule has been in recom-

mended use since its original publication; it has not, to our knowledge, been checked against an independently logged count of the calls a reward-bearing flyer actually receives.

Two related questions follow from this gap. The first is whether stating a reward at all — independent of amount — changes how many tip-off calls a flyer receives, relative to an otherwise identical flyer that mentions none. The second is whether the Schedule's assumed dosage-response curve, in which call volume is expected to climb steadily across all four tiers, holds up once real call logs are checked against it, or whether the relationship saturates earlier than the Schedule assumes.

We report a six-week matched-flyer field trial addressing both questions directly, with three contributions, each hedged to the evidence gathered:

1. We present a matched-pair field method for testing whether a stated reward, independent of its amount, is associated with a change in a lost-pet flyer's tip-off call volume, deployed across 35 residential streets over a full six-week posting window.
2. We benchmark the Neighbourhood Noticeboard Reward-Tier Schedule's assumed dosage-response curve against independently logged call volume across its four recommended tiers, finding the observed relationship consistent with the Schedule's assumption only below its midpoint.
3. We report an ablation directly comparing the Schedule's two highest tiers, finding no reliable difference in call volume between them, and discuss what this suggests for the Schedule's upper range.

Related work

Research on how owners actually recover a lost pet finds neighbourhood-level search — posting signs, walking the streets, asking neighbours directly — among the methods both owners and finders report using most, alongside a pet's own tendency to return unassisted [1], [2], [3], a pattern a larger national survey situates within markedly higher recovery rates for dogs than for cats [4]. Whether attaching a price to that search behaviour helps is contested rather than settled: a well-known field experiment found that a monetary fine for late nursery pickups raised, rather than lowered, the priced behaviour [5], a large observational dataset finds the reward-size-to-response relationship consistently S-shaped rather than linear [6], theoretical work describes extrinsic rewards displacing the reputational motives that would otherwise sustain a prosocial act [7], trust-game evidence decomposing cooperation into risk expectations, unconditional kindness, and reciprocity offers one account of why a reward's effect on a tip-off call needn't scale simply with its size [8], and a field experiment on reporting peers' wrongdoing finds monetary incentives can backfire once the reported behaviour is morally weighted [9]. The nearest analogue for informal street signage's own behavioural effect is the lawn-sign literature, which finds a real but modest vote-share effect with spillover to untreated neighbours [10], alongside the wider literature on how visible normative and environmental cues shape passersby behaviour [11],

[12], [13]. Measurement scholarship supplies the sharper warning for a schedule built on an unchecked dosage-response curve: Strathern's and Campbell's accounts of a proxy substituting for the behaviour it was meant only to track [14], [15], and recent work formalising the distinct ways a Goodhart-style target fails once optimised against [16], [17]. Elsewhere at this institution, a matched-pairs quasi-experiment found that naming a directly responsible individual on shared marker caddies raised staff-reported restocking confidence far more than it raised the caddies' actual stocked rate [18], and a validation of a predictive stationery-reorder forecast found it statistically indistinguishable from the reactive threshold it was meant to improve on [19]; both anticipate the gap between a schedule's assumed and observed behaviour reported below.

Method

Setting. We recruited 35 residential streets in the University's host municipality via a letterbox call for participation, screening for streets with at least one existing community noticeboard or a habitual power-pole posting spot. Each participating street hosted a matched pair of flyers for the same lost cat — a small brown tabby, described on both flyers as answering to “Tof-fee,” missing since a fixed date — identical in photograph, description, layout, and typeface. The two flyers in a pair differed in exactly one respect: one stated a reward amount, drawn from one of the Schedule's four recommended tiers (\$20, \$50, \$100, \$200); the other stated no reward at all. Each street's tier assignment was randomised in advance and stratified to give each tier eight or nine streets (nine at \$20, nine at \$50, nine at \$100, eight at \$200; 35 in total).

Call attribution. Both flyers in a pair carried the same intercept phone number, staffed by an automated system rather than a person, so that flyer wording rather than call-handling was the only thing that varied between them. Each flyer additionally printed a distinct three-digit reference code in the same type and position and asked callers to key it in on connecting; the system logged every call's timestamp against the code entered. Reference-code entry was successful on 94% of connected calls; the remainder were attributed by cross-referencing timestamp against street and retained.

Duration and outcomes. Flyers remained posted, unaltered, for six weeks. We recorded, for every flyer, the total number of attributed calls across the window and the day, since posting, each call arrived. Our two outcomes are therefore: whether a flyer received at least one call in a given week (the street-week level, used for the reward-presence comparison across all 70 flyers) and the flyer's total call count across the full window (used for the tier-level dosage-response comparison, restricted to the 35 reward flyers).

Analysis. We model the weekly any-call outcome with a logistic regression clustering standard errors by street, and the reward flyers' six-week total call counts with a Poisson regression on log reward amount, adapting standard dose-response curve-fitting conventions from toxicology and ecotoxicology to a behavioural outcome [20], [21], [22]:

$$\log(\mu_i) = \beta_0 + \beta_1 \cdot \log(r_i)$$

where μ_i is flyer i 's expected six-week call count and r_i its stated reward amount. A flattening dosage-response curve implies β_1 shrinking toward zero above the tier at which the Schedule's assumption stops holding.

Results

Flyers stating any reward received at least one tip-off call in 82 of 210 street-weeks (39.0%), against 50 of 210 street-weeks for otherwise identical no-reward flyers (23.8%) — a 15.2 percentage-point gap (95% CI 6.4-24.0; $z = 3.40$, $p < .001$). The gap held in the same direction across every one of the six posting weeks studied, though its size varied week to week.

Restricting to the 35 reward flyers, mean six-week call volume rose from 2.1 at the \$20 tier to 3.4 at the \$50 tier, then held essentially flat at 3.6 at the \$100 tier and 3.5 at the \$200 tier. The Schedule's own published table predicts a continued climb across the same range — 2, 4, 7, and 12 calls at the four tiers respectively — and the two curves diverge sharply above \$50 (Figure 1, Table 1). The fitted Poisson model put the reward-amount slope among reward flyers at $\beta_1 = 0.31$ (95% CI -0.04 to 0.66), consistent with a real but small association once amount alone — rather than reward presence — is isolated, well short of the near-linear response the Schedule's table assumes.

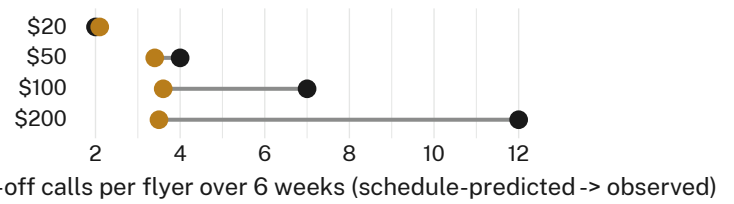


Figure 1: The Neighbourhood Noticeboard Reward-Tier Schedule's predicted call count climbs steeply across its four tiers; the observed six-week mean plateaus after \$50.

Tier	Schedule-predicted calls	Observed mean calls
\$20	2	2.1
\$50	4	3.4
\$100	7	3.6
\$200	12	3.5

Table 1: The Schedule's predicted call count per tier against the observed six-week mean.

Call-arrival timing tells a more uniform story than call volume does. Across all five arms — no reward and each of the four tiers — calls arrived in a similarly front-loaded pattern, concentrated in the first ten to fourteen days after posting and tailing off by the fourth week (Figure 2). Whatever a stated reward changed, it did not change when, in the posting window, a passer-by decided to call.

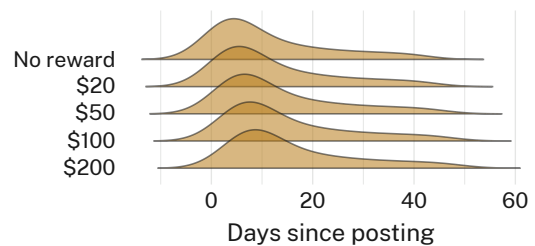


Figure 2: Call-arrival timing is near-identical in shape across the no-reward arm and all four reward tiers; only the overall volume differs.

An ablation comparing the Schedule's two highest tiers directly — \$100 against \$200 — found no reliable difference in six-week call volume between them (mean 3.6 versus 3.5; Welch's $t(14.9) = 0.18$, $p = .86$; Figure 3). Doubling the stated reward from \$100 to \$200 purchased, in this sample, no additional tip-off calls; we report the comparison in full rather than folding it into the tier-level result above.

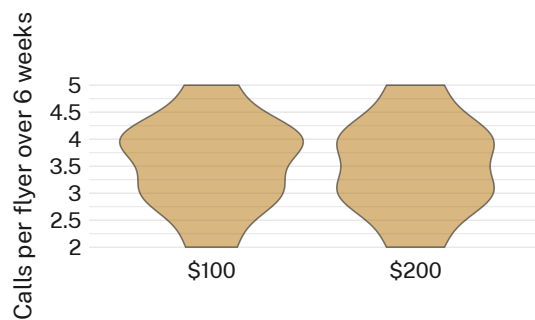


Figure 3: Doubling the stated reward from \$100 to \$200 left the distribution of six-week call counts per flyer unchanged ($p = .86$); we retain both tiers in the Schedule for completeness.

Discussion

We observe a reliable increase in tip-off call likelihood when a lost-pet flyer states any reward, and a real but modest further association between the reward's exact amount and total call volume once a reward is stated at all. Both readings sit inside the Schedule's founding assumption in one sense and against it in another: a reward helps, in the direction and rough order of magnitude the Schedule assumes, but the curve above its \$50 tier does not track the amount the Schedule recommends. Our results suggest the Schedule's lower tiers — \$20 and \$50 — capture most of the association reward presence and amount jointly carry, and that its two higher tiers, \$100 and \$200, may currently recommend a reward size the Schedule's own founding logic does not, on this evidence, purchase. We read the ablation's null result between the top two tiers as sharpening this reading rather than complicating it: if amount alone were doing the work the Schedule credits it with, doubling it should have moved the number, and it did not. Whether the same plateau appears outside a lost-pet flyer — for the Schedule's other covered noticeboard categories — is a natural next test, and one our single validation cannot answer on its own.

Limitations

Our trial covers 35 streets in a single municipality over a single six-week window, a modest sample for a four-tier comparison; the consistency of the plateau across both the tier-level comparison and the direct \$100-versus-\$200 ablation, however, suggests the pattern is unlikely to be an artefact of any single street or week. Call attribution relies on callers correctly

keying a printed reference code, successful for 94% of connected calls; the reassignment procedure for the remainder was manual and could not be independently audited. We studied one species, one flyer design, and one season; a schedule intended to generalise across all noticeboard postings, however, is the party whose generalisation claim this trial exists to test, not ours.

Conclusion

A reward-tier schedule can recommend a steadily climbing reward for a steadily larger search radius and still be right about only the lower half of its own range. Stating a reward at all raised the weekly likelihood of a tip-off call by fifteen percentage points over an otherwise identical flyer; raising the stated amount from \$20 to \$50 raised call volume further, and raising it again to \$100 or \$200 did not. Future work might extend the audit below \$20 and above \$200, and to the noticeboard categories — lost items other than pets, found-property notices — the Schedule also covers but this trial did not test.

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